Vatsal Ramanuj

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IIT Madras Electrical Engineering graduate (GPA: 9.29/10), currently pursuing an MSc in Communications and Signal Processing at Imperial College London. Combines strong quantitative and statistical foundations with Python-based data analysis, time-series modelling and backtesting. Experienced in evaluating trading signals, statistical relationships and strategy performance, with research interests in systematic trading, market microstructure and data-driven decision-making in financial and commodities markets.

Education

Imperial College, London
MSc in Communications and Signal Processing, Dept. of Electrical Engineering — 2026 - 2027

Indian Institute of Technology, Madras, India (IITM) — **9.29/10** GPA
B.Tech, Electrical Engineering, with a minor in Physics — 2022 - 2026

Projects

Cointegration-Based Pairs Trading Research — github.com/vatsalramanuj/CointegrationTesting
- Screened **5,000+** stock and ETF pairs with **Engle–Granger** tests under Bonferroni and Benjamini–Hochberg **FDR control**; evaluated mean-reversion strategies with train/validation/test splits, **walk-forward backtests** and transaction-cost sensitivity.
- Found cointegration strength to be almost uncorrelated with a pair's Sharpe ratio and returns, and no robust out-of-sample alpha after controlling for selection bias.

Limit Order Book Simulator and Market-Making Research — github.com/vatsalramanuj/Lobsim
- Built a price-time-priority matching engine in Python; replaying LOBSTER AAPL data (21 Jun 2012, **301,587 messages**) reproduces book levels 1–4 with **99.70% accuracy** (99.94% on engine-tracked messages). Throughput: **∼780,000 operations/s** (**∼26×** faster than the reference engine).
- Developed an event-driven simulator with latency, informed order flow and an **Avellaneda–Stoikov** market maker; across 12 paired seeds, achieved mean PnL of **2,415**, mean absolute inventory of **4.5**, maximum drawdown of **87**, and PnL/volatility of **15.4** at $\gamma = 0.01$.

Research and Technical Experience

ML based multistatic radar object localisation — Supervisor: **Prof. K Giridhar**
Bachelors Project — Sept'25 - May'26
- Developed a **physics-informed GRU–Transformer** that fuses TDOA/FDOA measurements to jointly estimate 3D position and velocity of up to 20 drone-sized targets.
- Trained on synthetic data using a **Singer motion model** and **Swerling-1 fluctuating targets**, with a composite loss for detection and regression.
- Achieved **∼14 m 3D localisation RMSE** under realistic scenarios and measurement noise while having a **near-constant inference time** much faster than classical approaches.

OLA Krutrim — Bangalore, India
Software Development Intern — June '25 - Aug '25
- Implemented and tested C++ modules for **advanced driver-assistance systems**, covering forward-collision warning, blind-spot detection and lane-departure warning.
- Set up **Docker-based deployment**, software-in-loop and hardware **integration tests**, and gcovr coverage tracking to 97% line/decision coverage.

SilverTouch Technologies Limited — Ahmedabad, India
ML Engineer — Dec'23 - Jan'24
- Applied ML and DL models like **LSTM, LightGBM, GRU**, etc, for weekly dengue-case forecasting across two South American cities, using historical time-series data to model temporal patterns.

Skills
- **Programming**: Python, C++, MATLAB; NumPy, pandas, SciPy, scikit-learn, PyTorch, TensorFlow, Matplotlib
- **Quantitative Methods**: Probability, Linear Algebra, Statistical Signal Processing, Time-Series Forecasting, ML
- **Tools**: Git, Docker, Linux, LaTeX, Mathematica

Achievements
- **National Champion, SWANotchFM** (IIT Kanpur): led the national champion team. 2025
- **JEE Advanced All-India Rank 1685** (top 1.1% of 150,000+) and **JEE Main Rank 623** (top 0.06% of 1M+), India's national engineering entrance exams. 2022
- **KVPY Fellowship** (Indian Institute of Science): national rank 228 of 40,000+ applicants. 2022
- **Aruna Lal Scholarship** (Physical Research Laboratory, Dept. of Space): 5 awardees in the state per year. 2021

Leadership Experience
- **Head, Horizon (Physics & Astronomy Club)**: led 50+ members, and organised 20+ talks and events with 100+ attendance.